ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	65.0	**Gender**	Male
Height (cm)	170.0	**Weight (kg)**	110.0
BMI	38.0	**Product Info 2**	D3
Occupation	Construction	**Annual Income**	$45,000.00
Insurance Type	Life	**Coverage Amount**	$200,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 8 of 8
Risk Category	**High Risk**
Recommended Annual Premium	**$40,000.00**

3. AI Actuarial Analysis

Actuarial Report: Life Insurance Application Evaluation

Applicant: HighRisk Customer
Date: October 26, 2023
Product: Life Insurance, Coverage: $200,000

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1. Risk Assessment

The applicant presents an exceptionally high-risk profile due to a confluence of adverse factors:

* **Age (65):** Advanced age significantly increases mortality risk.
* **BMI (38.0):** Severely obese, indicating substantial health complications risk (cardiovascular disease, diabetes, etc.).
* **Smoker (1):** Active smoking is a primary driver of premature mortality and dramatically elevates premiums.
* **Occupation (Construction):** This occupation is classified as hazardous, posing higher risks of accidents, injury, and occupational diseases, particularly at age 65.
* **Previous Claims (2):** A history of two previous claims is a significant red flag, suggesting a higher propensity for claims or underlying health issues that have materialized in the past.
* **Family History (1):** Indicates a genetic predisposition to certain conditions, further increasing risk.
* **Exercise (0):** Sedentary lifestyle exacerbates the health risks associated with obesity and age.
* **Alcohol (2):** Moderate alcohol consumption is noted but is a lesser concern compared to other factors.
* **Coverage Amount ($200,000) vs. Income ($45,000):** The coverage amount is approximately 4.4 times the applicant's annual income, which is relatively high for an individual with such elevated risk factors.

Overall, this profile represents a very high mortality risk, placing significant strain on the insurer's claims reserves.

2. Comparative Analysis

The current application, assigned a Risk Class of 8 with a predicted premium of $40,000, stands at the extreme end of the risk spectrum compared to the provided historical cases:

* **POL-2626 (customer_9):** With a Risk Class of 2 and a premium of $12,000, this case is significantly less risky. The current applicant's profile (age 65, BMI 38, smoker, previous claims 2) is drastically different, likely encompassing multiple severe risk factors absent in POL-2626.
* **POL-25755 (customer_4):** This case, with Risk Class 5 and a premium of $20,000, is also considerably lower risk than the current applicant. While customer_4 might have presented some elevated risks, they were clearly not as severe or numerous as those observed in "HighRisk Customer".
* **POL-31852 (customer_5):** This is the most similar historical case with Risk Class 7 and a premium of $30,000. While sharing a high-risk profile, "HighRisk Customer" is classified even higher (Risk Class 8) and commands a $10,000 higher premium. This suggests that the current applicant possesses an even more detrimental combination of risk factors (e.g., higher BMI, smoking status, more previous claims, or greater age) than customer_5. The low similarity scores (all below 43%) for these historical cases highlight the uniqueness and extremity of the current applicant's risk profile.

3. Anomaly & Fraud Detection

Several anomalies and potential red flags are present:

* **Extremely High Risk Class (8) with Low Prediction Confidence (33%):** The machine learning classifier identifies the applicant as the highest possible risk class but expresses very low confidence in this prediction. This indicates that while the model recognizes the extreme risk, it struggles to precisely categorize or price such an unusual profile, potentially due to a lack of similar historical data points for confident classification. This is a critical anomaly requiring manual intervention.
* **Combination of Severe Risk Factors:** The concurrent presence of advanced age, severe obesity, active smoking, and a hazardous occupation is a highly unusual and problematic combination for life insurance underwriting.
* **High Number of Previous Claims (2):** A history of two claims raises concerns about chronic health issues or a pattern of claiming behavior, which requires detailed investigation.
* **High Coverage Amount Relative to Income:** A coverage ratio of 4.4x income, coupled with the applicant's extreme risk profile, could indicate potential moral hazard or speculative intent. The financial justification for such a high coverage amount needs thorough verification.

4. Premium Recommendation

The calculated annual premium of $40,000.00 reflects the applicant's exceptionally high-risk profile.

* **Justification:** The combination of advanced age (65), severe obesity (BMI 38), active smoking, hazardous occupation (Construction), multiple previous claims, and a sedentary lifestyle collectively drives mortality risk to its highest level. The ML classifier assigning the highest Risk Class (8) is consistent with this assessment. Compared to the closest historical high-risk case (POL-31852, Risk Class 7, $30,000 premium), the $40,000 premium for Risk Class 8 is directionally justified by the incremental increase in perceived risk.
* **Concern:** Despite the directional justification, the extremely low prediction confidence (33%) from the ML model for this premium indicates significant uncertainty in the model's output for this particular applicant. This low confidence means the $40,000 premium, while appearing logical given the risks, might not be a robust or precise estimate from the automated system. It suggests the model is pushing the boundary of its reliable prediction range.

5. Underwriting Decision Recommendation

Recommendation: Refer for Manual Review with Extreme Caution

This application cannot be automatically approved or rejected based solely on the current data and ML output.

Justification:
1. **Extreme and Multifaceted Risk Profile:** The applicant presents an unparalleled combination of severe risk factors that demand expert underwriting assessment.
2. **ML Model Uncertainty:** The critically low prediction confidence (33%) in the highest risk class and premium indicates that the automated system itself is not confident in its assessment. Relying on such an output for a final decision carries significant risk of mispricing or inappropriate decision-making.
3. **Anomalies and Red Flags:** The high previous claims history, disproportionate coverage amount relative to

income, and the overall extreme health profile necessitate a thorough manual investigation.

Actions Required for Manual Review:
* **Full Medical Underwriting:** Obtain comprehensive medical records, including an Attending Physician Statement (APS), recent medical examinations, and laboratory results to fully assess current health status and severity of conditions related to high BMI and smoking.
* **Claims History Verification:** Investigate the nature and details of the two previous claims.
* **Financial Underwriting:** Verify the financial justification for the $200,000 coverage amount relative to income and dependents' needs.
* **Occupational Assessment:** Evaluate the actual physical demands of the "Construction" role at age 65, and consider potential for work-related disability or mortality.
* **Expert Actuarial Review:** An expert actuary should review all gathered information to determine if the risk is insurable at any price, and if so, to calculate a manually determined, robust premium, potentially considering non-standard policy terms or exclusions.

It is highly probable that, even with a manual review, this application may result in a decline or an even higher premium than currently calculated, should the risk be deemed insurable at all.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **$40,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.